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Methodology

Using autonomous recording units for vocal individuality: insights from Barred Owl identification

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ABSTRACT. Recent advances in acoustic recording equipment enable autonomous monitoring with extended spatial and temporal scales, which may allow for the censusing of species with individually distinct vocalizations, such as owls. We assessed the potential for identifying individual Barred Owls (*Strix varia*) through detections of their vocalizations using passive acoustic monitoring. We placed autonomous recording units throughout the John Prince Research Forest (54°27' N, 124°10' W, 700 m ASL) and surrounding area, in northern British Columbia, Canada, from February to April 2021. The study area was 357 km² with a minimum of 2 km between the 66 recording stations. During this period, we collected 454 Barred Owl calls, specifically the *two-phrase hoot*, from 10 recording stations, which were of sufficient quality for spectrographic analysis. From each call, we measured 30 features: 12 temporal and 18 frequency features. Using forward stepwise discriminant function analysis, the model correctly categorized 83.2% of the calls to their true recording location based on a 5-fold cross validation. The model showed substantial agreement between the recording station that the call was classified to originate from, and where the call was actually recorded. The most important features of the calls that enabled discrimination were call length, interval between the 4th and the 5th note, interval between the 6th and 7th note, and duration of the 8th note. Our results suggest that passive acoustic monitoring can be used not only to detect presence/absence of species but also, where vocalizations have individually distinct features, for population censusing.

Utilisation d'enregistreurs pour la reconnaissance vocale individuelle : aperçu de l'identification de la Chouette rayée

RÉSUMÉ. Les progrès récents en matière d'équipement d'enregistrement sonore permettent un suivi autonome à des échelles spatiales et temporelles étendues, ce qui peut permettre le recensement d'espèces dont les vocalisations sont distinctes, comme les chouettes et les hiboux. Nous avons évalué le potentiel d'identification de Chouettes rayées (*Strix varia*) individuelles par la détection de leurs vocalisations à l'aide d'un suivi acoustique passif. Nous avons installé des enregistreurs automatiques dans la forêt de recherche John Prince (54°27' N., 124°10' W., 700 m ASL) et ses environs, dans le nord de la Colombie-Britannique, au Canada, de février à avril 2021. L'aire d'étude couvrait 357 km² et il y avait un minimum de 2 km entre les 66 stations d'enregistrement. Au cours de cette période, nous avons recueilli 454 cris de Chouettes rayées, en particulier le hullement en deux phrases, à partir de 10 stations d'enregistrement dont la qualité était suffisante pour faire une analyse spectrographique. Pour chaque cri, nous avons mesuré 30 caractéristiques : 12 temporelles et 18 de fréquence. Au moyen d'une analyse de fonction discriminante progressive, le modèle a correctement catégorisé 83,2 % des cris à leur véritable emplacement d'enregistrement sur la base d'une validation croisée 5 fois. Le modèle a montré une forte concordance entre la station d'enregistrement classée comme étant celle d'où le cri était provenu et l'endroit où le cri a été réellement enregistré. Les caractéristiques les plus importantes des cris ayant permis leur discrimination sont la longueur du cri, l'intervalle entre la 4e et la 5e note, l'intervalle entre la 6e et la 7e note et la durée de la 8e note. Nos résultats montrent qu'on peut utiliser le suivi acoustique passif pour détecter la présence ou l'absence d'espèces, et aussi, dans les cas où les vocalisations ont des caractéristiques distinctes, recenser les populations.

Key Words: *BirdNET; identification; owls; sound recordings; passive acoustic monitoring*

INTRODUCTION

The ability to identify individual animals has many conservation applications (McGregor and Peake 1998), including tracking individual movements and habitat use, estimating population sizes, studying site/mate fidelity, and analyzing behavior and social interactions. Traditionally, individual identification has involved marking animals with physical tags, such as GPS units, PIT tags, leg bands, ear tags, and transmitters. While these methods are well established for monitoring individuals, they require capturing animals, thereby leading to time-consuming

and somewhat invasive data collection methods. Alternatively, some species can be individuated by observing distinctive features on animals, such as scars (Langtimm et al. 2004), fur spot patterns (Jackson et al. 2006), tail marks (Swanepoel 1996), stripe patterns (Karanth 1995), and body coloration patterns (Monteiro et al. 2014). Because these features are naturally occurring variation, they can be used to track individuals without requiring physical capture. The use of natural variation can often be applied to a larger dataset than when individual tracking involves physical capture.

One promising application of tracking using naturally occurring individual variation is acoustic monitoring (Gilbert et al. 2002, Hartwig 2005). Vocal individuality has been noted in many animal taxa, such as wolves (Sadhukhan et al. 2021), lions (Trapanotto et al. 2022), dolphins (Janik 2009), and owls (Ural Owls *Strix uralensis* [Zhou et al. 2020], European Eagle Owls *Bubo bubo* [Grava et al. 2008], African Wood Owls *Strix woodfordii* [Delpont et al. 2002], Tawny Owls *Strix aluco* [Galeotti and Pavan 1991], Barred Owls *Strix varia* [Freeman 2000], Western Screech-Owls *Megascops kennicottii* [Tripp and Otter 2006], Ryukyu Scops Owls *Otus elegans* [Takagi 2020], Eastern Screech-Owls *Megascops asio* [Nagy and Rockwell 2012], Northern Saw-whet Owls *Aegolius acadicus* [Otter 1996, Holschuh and Otter 2005], Great Horned Owls *Bubo virginianus* [Odom et al. 2013], Pygmy Owls *Glaucidium passerinum* [Galeotti and Pavan 1991], Sunda Scops-Owl *Otus lempiji* [Yee et al. 2016], and Great Gray Owls *Strix nebulosa* [Rognan et al. 2009]). Decades of research have demonstrated the effectiveness of using acoustic monitoring to identify individual owls, given the stable and distinctive features of their vocalizations. Almost all these studies, however, involved researchers tracking and manually recording single birds. Recent advances in the use of autonomous recording units (ARUs) for surveying bird communities may offer the potential to census populations using individually distinct vocal markers.

The use of ARUs in avian monitoring is currently experiencing significant growth, given the advantages they offer: reduced environmental impact and the ability to subject audio-based data to quantitative analysis. Autonomous recording unit surveys require only labor for deployment, retrieval, and analysis, which results in lower labor costs compared to traditional point count surveys for coverage of the same number of surveying periods. Furthermore, the development of analytical tools such as algorithms for automatic species identification based on sounds has expanded the potential applications of ARUs in large-scale acoustic monitoring. One of the most advanced algorithms is BirdNET, a deep neural network-based bird sound classifier capable of identifying North American and European bird species by their sounds (Kahl et al. 2021). The development of ARUs combined with BirdNET may also provide an efficient method for acoustically monitoring individual birds.

Despite being proposed and acknowledged as an effective passive alternative to capture/tagging for censusing owls (Dale et al. 2022), applying ARUs to owl individual identification could present several challenges. The signals recorded by ARUs are often of poor quality compared to manual recordings because the static location of audio recorders results in many recordings of distant birds. Owl calls can propagate over long distances, but they suffer from signal degradation and interference from background noise as the distance from the source increases (Baptista and Gaunt 1994). This may reduce the feasibility of using ARU recordings for individual owl identification. To enable individual identification, signals must exhibit clear "features" that can be accurately measured to distinguish subtle differences between birds. Temporal features of owl calls have been effective in individual identification; they include internote interval (Odom et al. 2013, Zhou et al. 2020), call length (Tripp and Otter 2006, Odom et al. 2013), note duration (Tripp and Otter 2006, Odom et al. 2013, Zhou et al. 2020), and number of notes (Zhou et al. 2020). Other studies have also demonstrated the efficiency of using frequency features in individual owl identification, such as

mean frequency (Otter 1996) and minimum and maximum frequency (Odom et al. 2013). Temporal features, though, are subject to degradation through scattering when subjects are recorded at greater distances (Yip et al. 2017). Similarly, frequency-dependent attenuation with distance could compromise the ability to distinguish among individuals. However, whether these impede the ability of ARU-collected sounds to differentiate between owls to conduct population monitoring has not been tested. We deployed ARUs as part of a project that assessed landscape biodiversity, and detected sufficient numbers of Barred Owls during the surveys to test the potential for using ARUs for individual identification.

Barred Owls occur in mature mixed-wood forests across much of Canada; therefore, they are often used as an indicator species in forest management. Furthermore, the expansion of the Barred Owl's range has led to overlap with the closely related Spotted Owl (*Strix occidentalis*) in the Pacific Northwest, and has been the focus of a number of studies on potential conflict between the two species (Long and Wolfe 2019). Developing an effective tool for monitoring Barred Owls could help contribute to evidence-based management plans. The aim of this study was to assess the feasibility of using ARU recordings not just for presence/absence detection, but potentially for distinguishing between recorded individuals to achieve more accurate censusing. When reviewing our landscape-level detections, we found that Barred Owls, which have been shown to exhibit individual vocal variation (Freeman 2000), were detected at a number of stations, so we chose this species as the subject of analysis. Our specific objectives were to (1) use the advanced bird sound classifier BirdNET to screen for Barred Owl signals detected in our long-term monitoring ARU project, (2) identify the temporal and/or frequency features of Barred Owl calls from ARU recordings that provided the best individual distinction, and (3) calculate the accuracy of identifying Barred Owls using acoustic features and compare it to studies on other owl species that used targeted, individual recordings. Ultimately, this serves as a baseline study to demonstrate the feasibility of using ARUs as a long-term monitoring tool for individual species, such as Barred Owls.

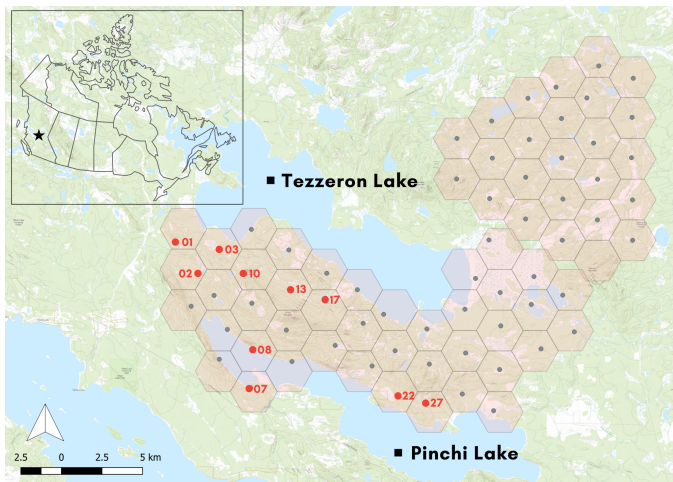
METHODS

Target species and study area

The Barred Owl is a non-migratory species that is widely distributed across North America. It is associated predominately with large tracts of mixed mature forests, often near water. Barred Owls are strongly territorial, a behavior proposed to result from limited nest sites (Mazur et al. 1998). Data from 158 banded Barred Owls revealed that the species' annual home range is typically less than 10 km² (Johnsgard 1988); however, movement is much more spatially restricted during the breeding season (February–April), when the average territory size is between 250 and 300 ha (Livezey 2007), representing a circular area with an approximately 1-km radius. The boundaries of home ranges are generally well maintained, and remain stable from year to year and even generation to generation (Nicholls and Warner 1972). Barred Owl vocal activity peaks during the breeding season, with highest rates of spontaneous calling occurring during late March to late April, prior to egg laying (Elderkin 1987). Another peak time occurs in the fall, potentially corresponding to the dispersal of the young.

The audio data were collected at the John Prince Research Forest (54°27' N, 124°10' W, 700 m ASL), British Columbia, Canada from February to April 2021, corresponding to the peak time of Barred Owl vocal activity. Autonomous recording units (AudioMoths [Hill et al. 2019]) were deployed at 66 sites throughout the study area as part of a larger biodiversity monitoring project; adjacent sites were at least 2 km apart. The sites were selected to include various forest age classes, compositions of forest, and harvesting activities in the region (Fig. 1), but many of the sites included suitable habitat for Barred Owls, which are known to occupy the region. Most sites with Barred Owl detections were concentrated in the western part of the study area, possibly due to the presence of favorable habitat characterized by mixed-wood forests. Conversely, the absence of detections in the eastern grid of the study area may have been due to recent forest harvesting (< 20 years), which would have limited Barred Owl settlement. However, a definitive explanation would require more detailed occupancy analyses.

Fig. 1. Locations of the 66 recording sites. All sites were located within or just outside the John Prince Research Forest, north of Ft. St. James, British Columbia, Canada: 42 sites were within the research forest, and 24 sites were northeast of the research forest. Dots represent the locations of recorders, and hexagons were superimposed to ensure spatial independence. Sites where Barred Owls were detected are marked in red, with site ID annotated.



The biodiversity monitoring project deployed ARUs in a hexagonal grid because the design is straightforward when considering the nearest neighborhood (Birch et al. 2007). This also facilitated sound analysis and spatial separation of sites needed for monitoring Barred Owls. Our deployment distance (i.e., 2 km) ensured that owls detected at one recording site would not be simultaneously recorded at a neighboring site, and the recording sites were separated by approximately twice the average radius of a Barred Owl pair's breeding home range (Livezey 2007). The ARUs were placed at the centroid of the hexagons unless there were logistic constraints (e.g., waterbodies, inaccessible terrain) in reaching those locations; in those cases, the ARUs were placed as close to the centroids as possible.

Audio data collection

We used AudioMoth (v1.1.0) ARUs to conduct audio recordings. The recording window was set from 6 PM to 10 PM daily (Pacific Standard Time); sunset time varied between 5:30 and 6:30 PM during the study period. The recording window aligned with the known peak in vocal activity of Barred Owls during the breeding season (0–2 hours after sunset) (Clement et al. 2021). Each AudioMoth was programmed to record actively for 10 minutes, followed by a 20-minute period of no recording. This recording cycle was repeated throughout the 4-hour recording window, which resulted in a total of eight 10-minute recordings collected per day, per site. At the end of the project, approximately 20,000 10-minute recordings were collected, representing approximately 3000 hours of monitoring effort. A more extensive recording effort was needed during nocturnal surveys for owls compared to typical songbird monitoring programs due to the lower population densities of most large owls and because vocal bouts occurred more infrequently than songbird vocalizations. All recordings were digitized at a sampling rate of 48 kHz and were stored as 16-bit sound files (wav format) for subsequent analysis. The high sampling rate ensured accurate representation of the owl vocalizations and preserved the necessary frequency range for detailed examination.

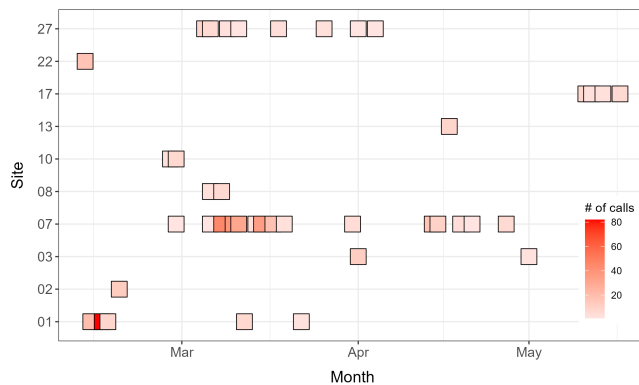
Sound feature processing

Following data collection, we employed BirdNET (Kahl et al. 2021) to identify recordings that contained Barred Owl vocalizations. Specifically, we obtained the source code of BirdNET through the GitHub repository (<https://github.com/kahst/BirdNET-Analyzer>) and systematically processed all of our data through BirdNET in the High-Performance Computing lab at the University of Northern British Columbia. We provided BirdNET optional parameters to enhance species detection. We specified the recording longitude and latitude based on the site location and the week of the year based on the recording date. BirdNET analyzes sound recordings by 3-second segments and produces outputs as .csv files, with columns indicating the detected species and a “detection confidence” ranging from 0.1 to 1.0. Higher detection confidence values indicate a greater certainty associated with a species detection. The raw output from BirdNET indicated there were approximately 2500 3-second segments in our ARU recordings that had detection confidence greater than 0.1 for Barred Owl vocalizations (i.e., more than 2500 3-seconds segments potentially included Barred Owl vocalizations). We then manually inspected all the recording segments that BirdNET indicated as potentially including Barred Owl vocalizations; from this pool, most incidences were true-positive Barred Owl detections but included a variety of call types, and some were too faint for spectrographic analysis.

Our field recordings captured various types of Barred Owl vocalizations, including the *ascending hoot* and *two-phrase hoot* (Mazur and James 2021). The *ascending hoot* has 6–9 regularly spaced notes followed by a downwardly inflected *hoo-aw* (Audio 1). However, the most frequently recorded vocalization is the *two-phrase hoot*, consisting of two groups of four or five notes each, and commonly being phoneticized as *Who cooks for you, who cooks for you all?* (Audio 2). The *two-phrase hoot* was suggested to be a territorial call (Johnsgard 1988) and a contact call

(Brewster and Chapman 1891). This vocalization was made throughout the night and was produced during spontaneous bouts of vocal activity and when an owl confronted conspecifics (Odom and Mennill 2010a). This hoot can be made by both sexes and occasionally by juveniles, although juveniles sometimes omit a few notes at the end. Further, the *two-phrase hoot* routinely occurred both inside and outside of duets (Odom and Mennill 2010b). Given that the *two-phrase hoot* was the dominant call type detected in our study area and can be used to identify individuals (Freeman 2000), we selected recordings of it (hereafter “Barred Owl calls”) as our target for the subsequent analysis. In total, 454 calls from 10 sites were of sufficient amplitude and clarity to permit spectrographic analysis and were included in the subsequent analyses; each site yielded 9–217 calls (Fig. 2). Calls with sufficient quality were defined as those with a signal-to-noise ratio above 4 dB.

Fig. 2. Barred Owl calls recorded during the 2021 breeding season. In total, 454 calls from 10 sites were analyzed.

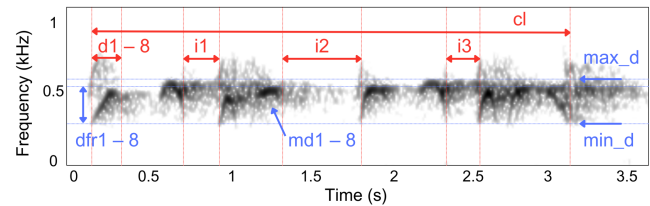


The recordings were processed using Raven Lite (K. Lisa Yang Center for Conservation Bioacoustics at the Cornell Lab of Ornithology 2023), a free software for spectrogram notations. We imported the recordings that contained Barred Owl calls into Raven Lite software, manually inspected them, and created the selection table (i.e., boundaries for signals of interest) of each Barred Owl call. For this, spectrograms were generated using the Raven Lite default setting Hanning Window with window size 256 samples, which resulted in a frequency resolution of 90Hz and a temporal resolution of 10 ms. Selection tables produced by Raven Lite defined the boundary of each Barred Owl call (i.e., start time and end time of the call; maximum and minimum frequency of the call).

To measure the sound features within each selection table, such as dominant frequency and duration of the call, we used the R (version 4.3.2 [R Core Team 2024]) package warbleR (Araya-Salas and Smith-Vidaurre 2017), a free modular tool for sound analysis and synthesis, which uses Raven Lite selection tables as input. Specifically, we used the *spectro_analysis()* function to extract 12 temporal features and 18 frequency features from each Barred Owl call (Fig. 3).

These 30 features were selected because they are standard measurements used in vocal individuality studies of non-passerines (e.g., Holschuh and Otter 2005, Tripp and Otter 2006).

Fig. 3. Spectrogram of a Barred Owl call, consisting of eight notes. We measured 12 temporal features (marked in red) and 18 frequency features (marked in blue). *cl* is the total length of the call, *d* is the duration of individual notes within the call, *i* is the interval between notes, *dfr* is the range of dominant frequency in individual notes, and *md* is the mean dominant frequency in notes. *max_d* and *min_d* are max and min dominant frequency of the call, respectively.



The 12 temporal features, measured in seconds, were the total length of the call (*cl*), the duration of individual notes within the call (*d1–d8*), the interval between the 2nd and 3rd note (*i1*), the interval between the 4th and 5th note (*i2*), and the interval between the 6th and 7th note (*i3*). The 18 frequency features, measured in Hz, encompassed the range of dominant frequency of individual notes (*dfr1–dfr8*), the minimum dominant frequency of the call (*min_d*), the maximum dominant frequency of the call (*max_d*), and the mean dominant frequency of the notes (*md1–md8*).

Statistical modeling

We used linear discriminant analysis (LDA) to classify sounds collected from individual sites. Linear discriminant analysis is a widely employed statistical technique for classifying unique groups using a linear combination of features that maximizes the discrimination between different groups within a dataset (Xanthopoulos et al. 2013). We used sound features, specifically temporal and frequency features, as predictor variables, and individual site ID as the response variable in the LDA. The LDA models were fitted using the *lda()* function in R package MASS (Venables and Ripley 2002).

For variable selection, we used a stepwise forward approach based on Wilks’ lambda statistic, a statistical measure commonly used in LDA to quantify the importance of variables in a model. This was done using the *greedy.wilks()* function in R package klaR (Weihs et al. 2005). Initially, the model was defined by selecting one variable that exhibited the greatest separation between groups. Subsequently, additional variables were incorporated based on their Wilks’ lambda statistic. This iterative process of adding variables continued until there was no further improvement in the model’s performance, as assessed by overall accuracy and kappa statistics. The overall accuracy and kappa statistics were derived by the average performance of a 5-fold cross-validation process, where the dataset was randomly divided into an 80% training dataset and a 20% test dataset. We used a bootstrap approach with 100 iterations of the 5-fold cross-validation process to compute the standard error for both overall accuracy and kappa statistics.

Spatial distance and vocal similarity

Even though the recording sites were at least 2 km apart, there was still potential that the same Barred Owl could be detected at two adjacent sites on sequential nights if individuals moved extensively within or between nights (e.g., movement distances exceeded the inter-recorder spacing such that the bird could be recorded at neighboring stations at different times in the recording season). As a result, vocalizations recorded at neighboring sites could appear to overlap in temporal/frequency characteristics (i.e., they would have similar multivariate centroids in the LDA model), and the model could misclassify calls recorded at one location as having originated at a neighboring location. If Barred Owls were moving extensively between recording sites, we would predict that at sites with closer spatial distances, Barred Owl calls that had more similar vocal features would be detected. To examine this effect, we measured the correlation between spatial distance and the vocal similarity of Barred Owl calls at each site. Spatial distances were determined by measuring the physical distances between the recording sites where the Barred Owls were detected. Vocal similarity was assessed by calculating the Euclidean distance between the multi-dimensional centroids of linear discriminant components of the recorded owls. Measuring the Euclidean distance between the centroids is a means of determining the degree of similarity of two vocalizations in multivariate clusters.

RESULTS

Temporal and frequency features

Descriptive statistics for the 30 sound features extracted from the 454 Barred Owl calls are reported in Table 1. The focus of this examination was on the coefficient of variation (CV) for these features, considering two perspectives: CV across all sites (CV_all) and CV within each specific recording site (CV_site). Features that are indicative of individual variation should have lower CV_site relative to CV_all. Based on paired *t* tests, both frequency and temporal features showed significant differences between CV_all and CV_site ($\alpha = 0.05$), which suggests the potential for these features to contribute to distinguishing sounds collected from different sites.

The stepwise forward variable selection method based on the Wilk's lambda criterion determined the most influential variables for distinguishing the calls recorded from individual locations. The top 10 variables, in descending order, were total length of the call (cl), interval between the 4th and 5th note (i2), interval between the 6th and 7th note (i3), duration of the 8th note (d8), mean dominant frequency of the 6th note (md6), duration of the 3rd note (d3), mean dominant frequency of the 8th note (md8), duration of the 2nd note (d2), mean dominant frequency of the 5th note (md5), and mean dominant frequency of the 1st note (md1). It is noteworthy that the initial four variables selected in the model were temporal features, which highlights their greater discriminative capability in distinguishing individual Barred Owl vocalizations compared to frequency features.

Linear discriminant analysis model

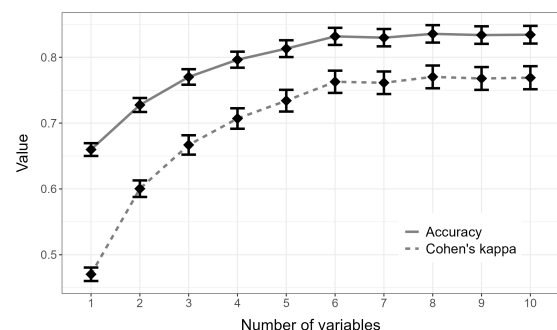
Variables were added incrementally to the cross-validated LDA model based on their discriminatory power in distinguishing between sites. As the number of variables increased in the model, there was an increase in both the overall accuracy and the kappa

Table 1. Mean, standard deviation (SD), and coefficient of variation (CV) for 30 sound features: 12 temporal features and 18 frequency features.

Feature type	Feature	Code	Mean	SD	CV all (%)	CV site (%)
Frequency	Range of dominant frequency in 1st to 8th note (Hz)	dfr1	229	76	33	31
		dfr2	222	70	31	29
		dfr3	217	70	32	32
		dfr4	189	69	37	37
		dfr5	232	74	32	26
		dfr6	221	80	36	34
		dfr7	219	67	30	32
		dfr8	232	72	31	28
Frequency	Maximum dominant frequency of the call (Hz)	max_d	627	40	6	5
Frequency	Minimum dominant frequency of the call (Hz)	min_d	322	58	18	15
Frequency	Mean dominant frequency of the 1st to 8th note (Hz)	md1	532	41	8	6
		md2	564	30	5	4
		md3	533	30	6	4
		md4	552	29	5	4
		md5	561	32	6	4
		md6	572	27	5	3
		md7	540	26	5	4
		md8	555	28	5	3
Temporal	Duration of the 1st to 8th note (sec)	d1	0.17	0.04	25	21
		d2	0.19	0.03	16	12
		d3	0.12	0.03	25	18
		d4	0.19	0.09	46	28
		d5	0.19	0.04	19	17
		d6	0.20	0.03	14	12
		d7	0.13	0.03	23	18
		d8	0.36	0.09	26	24
Temporal	Interval between the 2nd and 3rd note (sec)	i1	0.41	0.02	5	3
Temporal	Interval between the 4th and 5th note (sec)	i2	0.64	0.06	9	5
Temporal	Interval between the 6th and 7th note (sec)	i3	0.42	0.02	5	4
Temporal	Total length of the call (sec)	cl	3.06	0.14	5	3

statistics (Fig. 4). However, the performance of the model reached a point of saturation when six variables were included. Beyond that point, the incremental improvement in model performance by adding more variables was marginal. Consequently, we selected the first six features with the highest discriminatory power (i.e., cl, i2, i3, d8, md6, and d3) as our final model.

Fig. 4. Change in overall accuracy and kappa statistic when adding variables incrementally, with standard error derived by bootstrapping of 100 iterations. The performance of the model increased with the number of variables and reached a point of saturation when six variables were included.



The correlations between discriminant function and predictor variables (i.e., loadings) are presented in Table 2. The first two discriminant functions exhibited the highest correlation with the total length of the call (cl) and the interval between the 6th and 7th note (i3), respectively. Additionally, these top two discriminant functions collectively accounted for 80% of the total variability in the dataset (57% and 23%, respectively).

Table 2. Correlations between discriminant function and predictor variables (i.e., loadings). High correlation (absolute value ≥ 0.5) is shown in bold. The trace value reflects the among-group variation captured by that specific discriminant function. The number of functions is either N-1, where N is the number of groups, or the number of predictors, whichever is smaller. We had 10 groups and 6 parameters; thus, we had 6 linear discriminant (LD) functions from the linear discriminant analysis model.

LD functions	Sound features [†]						Proportion of trace
	cl	i2	i3	d8	md6	d3	
LD1	0.81	-0.48	0.33	0.63	0.67	-0.34	57%
LD2	-0.26	-0.71	-0.75	0.42	-0.10	0.23	23%
LD3	0.23	0.22	-0.43	-0.27	-0.23	0.46	12%
LD4	-0.26	-0.34	-0.08	-0.71	0.50	0.47	7%
LD5	-0.36	-0.14	-0.31	-0.45	0.15	-0.64	1%
LD6	-0.18	-0.29	0.21	-0.39	-0.46	-0.01	0%

[†] cl: total length of the call; i2: interval between the 4th and 5th note; i3: interval between the 6th and 7th note; d8: duration of the 8th note; md6: mean dominant frequency of the 6th note; d3: duration of the 3rd note.

We also conducted a principal component analysis and received comparable results to the discriminant function analysis (see Appendix 1 for details).

Identification performance

The final model achieved an overall accuracy of 83% and a kappa statistic of 0.76, which implies substantial agreement between the classification of calls into the predicted site (which recording site the model determined the call had originated from) versus observed site (the actual recording site where the call was recorded). The ordination plot of the final LDA model showed the distribution of each of the 454 Barred Owl calls in a reduced dimensional space (Fig. 5). The calls from the same site generally clustered closely together, indicating distinct groups, such as calls from site 01 (grey), site 07 (light blue), and site 17 (yellow). However, we also found some overlap of the clusters, indicating insufficient discrimination, such as between site 22 (dark blue) and all other sites.

The confusion matrix provides a concise overview of the performance achieved by our final model (Fig. 6A). This tabular representation offers a comprehensive count of calls assigned by the model to their predicted site of origin (predicted site), as compared to the sites where recordings were known to originate from (observed site). Most of the calls were correctly classified by the model—the predicted site aligning with the observed site (cells along the diagonal). However, there were instances of misclassification. Twenty-one calls from site 03 were misclassified as being from site 07, six calls from site 08 were misclassified as being from site 07, six calls from site 13 were misclassified as being from site 17, and six calls from site 22 were misclassified as being

Fig. 5. Ordination plot of the final linear discriminant analysis model. Only the first two discriminant functions are shown. Some sites have overlapping sound features, indicating insufficient discrimination.

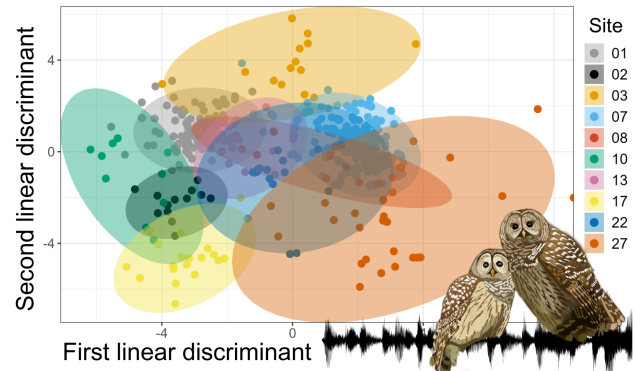
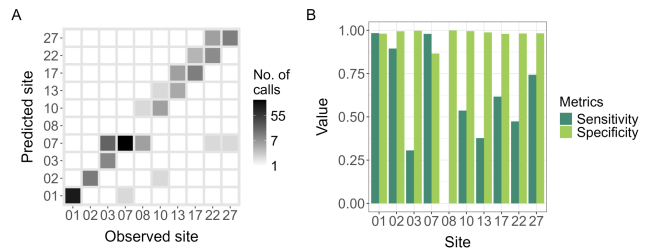


Fig. 6. (A) The confusion matrix summarizes the final linear discriminant model's outcomes, with call counts averaged from 100 bootstrap iterations. (B) Sensitivity and specificity calculated for each site.



from site 27. We computed the sensitivity (proportion of correctly identified positive instances) and specificity (proportion of correctly identified negative instances) for each site (Fig. 6B). The model exhibited high specificity (> 0.85) across all sites, while sensitivity varied from 0 (site 08) to 0.98 (site 01).

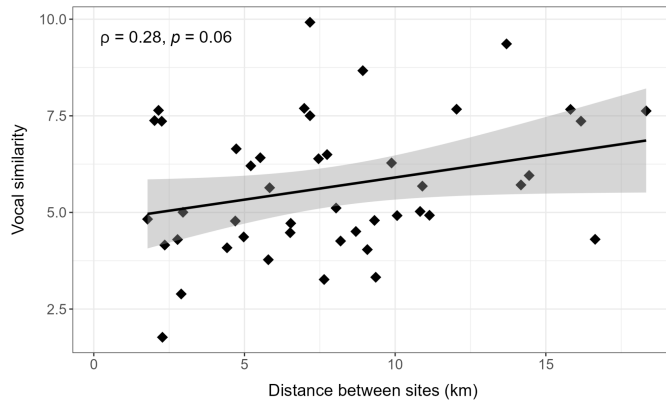
Correlation between spatial distance and vocal similarity

We examined the relationship between spatial distance (distance between any two recording sites) and vocal similarity (measure of Euclidean distance between the multidimensional centroids of the ordination plots as represented in Fig. 5) and found a correlation coefficient of 0.28 ($p = 0.06$), indicating a relatively low degree of association between spatial distance and vocal similarity (Fig. 7).

DISCUSSION

Our study offers evidence that supports the feasibility of employing individual acoustic monitoring using ARUs to identify individual birds, particularly in the context of our focal species, the Barred Owl. Despite inherent variability in audio recordings due to the fixed ARU placements, our analysis suggests that the

Fig. 7. The correlation analysis showed a low correlation (correlation coefficient = 0.28) between the distance and vocal similarity of Barred Owl calls.



discriminant ability of our models was not significantly compromised, which demonstrates that ARU recordings can capture sounds of sufficient quality to identify individuals. While our research did not involve a marked population of Barred Owls or directional recordings from individuals, our findings indicate promise for discriminating among individual birds based on vocalizations. We acknowledge the importance of replicating this work on populations of radio- or GPS-tracked individuals to validate the effectiveness of ARU-based acoustic monitoring in discriminating among individual birds.

Our LDA model achieved an accuracy of 83% in assigning calls to the recording sites, which is slightly lower than but still comparable to that of other studies that used manual sound recording, which achieves better control in recording quality (e.g., 89% for female European Eagle Owls [Grava et al. 2008], 99.1% for Tawny Owls [Galeotti and Pavan 1991], 97.4% for Ryukyu Scops Owls [Takagi 2020], 89% for Eastern Screech-Owls [Nagy and Rockwell 2012], and 92% for Western Screech-Owls [Tripp and Otter 2006]).

In our study, temporal features were the most powerful in distinguishing vocalizations of individual Barred Owls. This was unexpected, given that temporal features are mostly likely to suffer degradation with distance between the bird and the recorder (Bradbury and Vehrencamp 2011). Unlike manual recordings, ARUs introduce a challenge in establishing a standard distance-to-recording subject framework. Consequently, this discrepancy could complicate the precise measurement of the start and end of notes on spectrograms, owing to scattering effects. We predicted this might compromise the precision of temporal feature measurements, which could reduce the ability to show subtle differences among individuals. However, our LDA model demonstrated that, despite using static passive recorders, temporal measures still retained the greatest discriminant ability in distinguishing individual calls emanating from different sites. This finding corresponds well with Freeman's (2000) study of Barred Owl individual identification, which indicated that temporal features, duration, and interval of notes were powerful in distinguishing individuals.

We found a low correlation between the spatial distance between recordings sites where birds were detected and the vocal similarity of the Barred Owl calls recorded at each of the pair-wise ARU

stations. This suggests that it was unlikely that Barred Owls were moving between, and being recorded at, adjacent sites. As a result, the model's misclassification of calls between observed and predicted sites was likely not due to birds moving between and being detected at adjacent recording sites. This suggests that most of the Barred Owls were somewhat sedentary, at least within 2–3 km of the recording center, within a single breeding season. Therefore, species with restricted home ranges, such as the Barred Owl, are good subjects for using acoustic monitoring to census populations. With broad spacing of the recorders, large areas can be covered and monitored simultaneously, yet our data suggest that owls remained within their individual territories throughout the breeding season. Some of the model misclassification may have been due to more than one individual (e.g., both male and female within a pair) being recorded at a single site (Odom and Mennill 2010a). We observed instances of duet activity in specific study sites, indicating the presence of owl pairs within these areas. Duet calls were documented at site 15, 27 (Audio 3), and 28 (Audio 4). Previous research demonstrated that male and female Barred Owls can be distinguished based on their *two-phrase hoots* (Odom and Mennill 2010a), with females typically producing higher frequency calls compared to males. The identification of the sex of owls within the same site could potentially enhance the accuracy of individual identification models. Therefore, future research could explore whether ARU recordings within a site cluster sufficiently to differentiate between calls made by a single bird versus counter-singing pairs.

Given that the dataset used in this study was constrained to a single breeding season, an assessment of vocal individuality across multiple years was not investigated. Nonetheless, prior studies of owl vocalizations (e.g., Galeotti and Pavan 1991, Nagy and Rockwell 2012, Takagi 2020) consistently found strong stability of sound features over years based on manual recordings. To expand the applications of passive acoustic monitoring, we suggest that future studies use ARUs to investigate the constancy of owl vocalizations over extended periods. This can be done by collecting multi-year acoustic data at the same sites and integrating tracking techniques such as MOTUS or GPS to verify individuals. Upon method validation, passive acoustic monitoring enables prolonged observation of individual owls, which facilitates the study of behaviors such as year-to-year fidelity and territorial dynamics. The ultimate objective is to use passive acoustic monitoring as a tool for comprehensive investigations of individual owls.

CONCLUSION

This study establishes that the *two-phrase hoot* of Barred Owls, as captured by autonomous recording units, exhibits sufficient natural variation in sound features to potentially identify individual owls. While direct proof of individual identification by ARU is not provided, our research lays the groundwork for using passive recordings in owl censuses and individual monitoring. Our findings highlight the discriminatory power of temporal features in Barred Owl identification, despite their susceptibility to degradation with increasing distance from the recorder. Notably, the low correlation between spatial distance and vocal similarity revealed the stability of Barred Owl vocalization, both temporally (during the breeding season) and spatially (within a 2- to 3-km radius), which makes the species well-suited to acoustic census using monitoring techniques. Future studies could further

explore the use of this technique with other owl species, and investigate inter-year comparisons at the same sites, paired with tracking methods such as MOTUS or GPS to validate individuality. This approach would expand the scope of passive monitoring applications in studying individual owl behaviors.

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Data Availability:

All the analysis R script has been deposited in the GitHub repository: https://github.com/SunnyTseng/thesis_aru_owl_individual

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Appendix 1

We conducted a principal component analysis on the 30 sound features for 454 calls. We determined the number of principal components using broken stick method proposed by Jackson (1993) and selected the first 3 principal components (Table A1). These three components collectively explained 55% of the variance. Notably, the first component showed a higher correlation with frequency features, while the second and third components exhibited correlations with temporal features (Table A1; Fig. A1). The observed overlaps in the clusters align with expectations, as principal component analysis excels in reorienting original variables rather than distinctly separating clusters (Fig. A2).

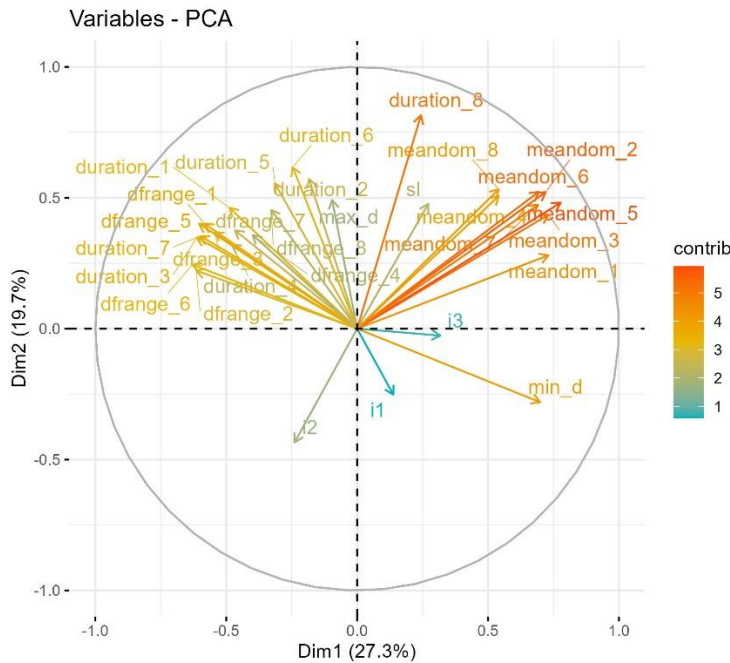


Fig. A1. PCA visualization of all the original variables on the PC1 and PC2.

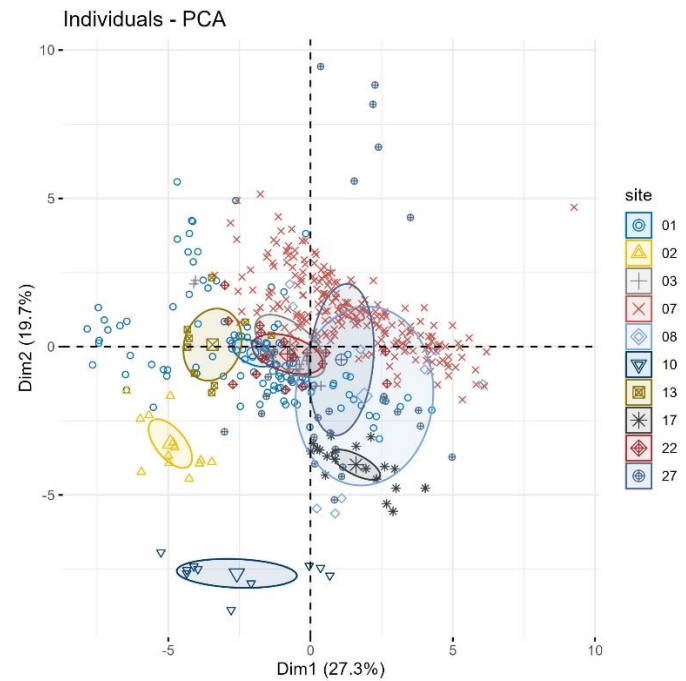


Fig. A2. PCA visualization of all the observations reprojected on the PC1 and PC2.

Table A1. Statistics of the PCA analysis on the 30 sound features.

Components	Variance explained	Accumulated variance	Broken stick value	Correlated variables (absolute cor > 0.6)
Comp1	27%	27%	0.13	min_d, md_1, md_2, md_3, md_5, md_6, md_7
Comp2	20%	47%	0.09	d_6, d_8
Comp3	8%	55%	0.08	cl, i1, i3